Tuning of bending and torsional modes of bars used in mallet percussion instruments

Filipe Soares, Jose Antunes, Vincent Debut

JASA ARTICLE

Tuning of bending and torsional modes of bars used in mallet percussion instruments

Filipe Soares,[1,a)] Jose Antunes,[1] and Vincent Debut[2]

[1]*Centro de Ciências e Tecnologias Nucleares, Instituto Superior Técnico, Universidade de Lisboa, Estrada Nacional 10, Km 139.7, Bobadela LRS 2695-066, Portugal*

[2]*Escola Superior de Artes Aplicadas, Instituto Politécnico de Castelo Branco, Avenida Pedro Álvares Cabral, n° 12, Castelo Branco 6000-084, Portugal*

ABSTRACT:

The present paper proposes a generic design-optimization procedure for the concomitant tuning of bending and torsional modes of bars used in mallet percussion instruments. The undercut model uses a series of discontinuous cuts aimed to facilitate the manufacturing process. Compared to one-dimensional beam models, the use of three-dimensional (3-D) finite element modeling not only allows for the calculation of torsional modes but also provides an increased accuracy in the prediction of modal frequencies, an important aspect when dealing with the demanding tuning tolerances required in a musical context. A global optimization problem is formulated and solved using a surrogate function algorithm, which enables fast computations even with the expensive function evaluations associated with 3-D finite element models. Modal identification of experimental bars demonstrates the potential of the proposed procedure, leading to bars with demanding tuning ratios (six target frequencies) at absolute tuning deviations typically below 15 cents. Measurements of the radiated sound from the experimental bars illustrate the benefits of the improved designs compared to those without torsional tuning. The proposed framework, aside from dealing with the comprehensive tuning of percussive musical bars, also accounts for important practical considerations regarding efficient optimization, modeling accuracy, and manufacturing complexity.
https://doi.org/10.1121/10.0006573

(Received 7 May 2021; revised 14 September 2021; accepted 17 September 2021; published online 14 October 2021)

[Editor: Andrew Morrison] Pages: 2757–2769

I. INTRODUCTION

Mallet percussion instruments, like the marimba or the vibraphone, are made up of a series of tuned bars that radiate sound when struck by a mallet. An important aspect of the instrument's intonation and timbre quality is related to the relationship between the various modal frequencies of the bars. The modal frequencies of a bar (beam) with uniform cross section are not harmonically related, a feature common to most tonal instruments. Hence, the tuning of idiophone bars is typically pursued via an extraction of material from the bar's underside such that its modal frequencies become harmonically aligned, as illustrated in Fig. 1.

At the present time, the design of the undercuts is mostly carried out by makers on an empirical basis, via trial and error. By far the most common undercuts found in commercial instruments are based on an arched shape, as illustrated in Fig. 1 (right). This typically aims to place the frequencies of the first three bending modes in harmonic relation, commonly referred to as "triple tuning." This design is pursued mostly for the low/medium register of the instruments, while in the higher register only the first two bending modes are typically tuned.

The variety of tuning ratios found on today's instruments is not very diverse. For vibraphones and marimbas, the frequency of the second mode is typically tuned to 4 times the fundamental (double octave interval), while the frequency of the third mode, although variable, tends to lie near 10 times the fundamental (triple octave plus a major third). In xylophones, the frequency of the second mode is commonly tuned to 3 times the fundamental, and the frequency of the third mode, although not always considered in the tuning process, typically lies between 6 and 10 times the fundamental.[1,2] It is likely that these traditional tuning ratios have evolved and become standardized due to the simplicity of the associated undercuts.

Nowadays, with the significant increase in computational power, efficient optimization methods, and accurate physical modeling techniques, as well as the emergence of CNC machining, there is a compelling argument to reassess the design of these instruments using a more generic and versatile engineering approach.

Indeed, several studies have been published over recent years on the optimal design of bars in mallet percussion instruments using various modeling approaches and optimization methods. Early works of Bork[3,4] investigated various features of mallet percussion instruments including the dynamics of a mallet stroke,[5] the acoustics of resonators, and, of course, bar tuning. He modelled the effect of

[a)]Electronic mail: filipedcsoares@gmail.com, ORCID: 0000-0003-3914-6026.

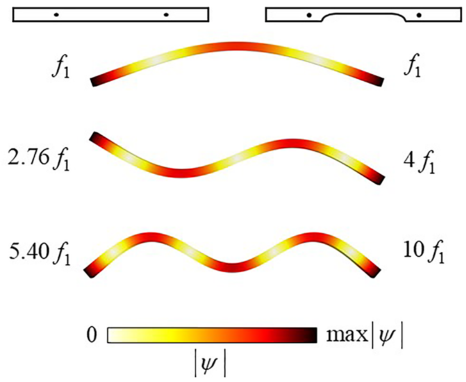

FIG. 1. (Color online) Three lowest vertical-bending mode shapes ψ and corresponding frequency ratios in a bar with uniform cross section (left) and a bar with a traditional undercut (right).

localized cuts on the frequencies of the lowest bending modes, providing a practical physics-based guide to tune the first three modes of (traditional) beams. Additionally, Bork and Meyer[6] investigated the subjective perception and tonal evaluation of xylophone beams with various tuning ratios. Orduña-Bustamante[2] explored an undercut based on a parabolic arc, similar to those found in traditional instruments, and was able to find geometries to various common tuning ratios.

Perhaps the first to explore beams with discontinuous profiles was Summers *et al.*,[7] who investigated the frequency of the two lowest bending modes in beams with a single rectangular cut, with variable height and length. Subsequently, Petrolito and Legge[8,9] have explored the tuning of bars with multiple profile discontinuities and showed the design potential of such geometries. In their model, the geometry of the undercut was defined by the heights of N beam subsections of equal length. This typically led to an optimization problem with few design variables (<10), whose objective function relied on the numerical solution of a one-dimensional (1-D) finite element (FE) model. Aside from reducing the optimization problem to few variables, the resulting optimized geometries can be easily manufactured using common milling machines, due to their simple rectangular profiles.

Henrique and Antunes[10] have devised a method that further reduces the variables of the optimization problem by developing a smooth undercut profile in terms of a set of orthogonal functions (e.g., Fourier or Chebyshev) and optimizing their amplitude coefficients. This method allowed the design of optimized shapes for demanding tuning ratios with a minimal variable set. The curvilinear nature of these undercuts could also have the advantage of being more robust to machining errors, compared to profiles with discontinuities. However, they would require more complex manufacturing procedures.

Despite these advances, tuning optimization of bars is generally addressed considering bending modes only, typically using simple one-dimensional models (Euler or Timoshenko beams). However, numerous reports[3,4,11–13] have pointed out the importance of torsional modes, which can have an adverse effect on the bar sound, particularly when their frequencies are close to those of the tuned bending modes. Moreover, the musical context requires a very low tolerance for errors in tuning, and the typically used 1-D models frequently fail to reproduce the frequencies of actual bars within acceptable tolerances. The work of Petrolito and Legge,[8] for example, used a 1-D Timoshenko beam model to tune the first three bending modes of undercut bars and commonly reported deviations of over 100–150 cents between modelled and measured modal frequencies in both aluminum and wooden bars. Consequently, the necessity of higher order models, with more accurate frequency predictions, is regularly suggested.[8–10,14]

Naturally, two-dimensional (2-D) plate models can be used to calculate torsional and bending modes at a relatively low computational cost. However, since modeling accuracy is a primary concern in a musical context, we opted to use three-dimensional (3-D) FE models, which allow the calculation of both bending and torsional modes without the risk of compromising modeling accuracy. This framework raises the question of computational efficiency and convergence of optimization schemes since the typical computation time of a function evaluation using a 3-D model increases significantly compared to simpler 1-D models. In the spirit of improving manufacturing practicality, here we use a discontinuous undercut model, defined by several rectangular cuts, whose lengths and heights are the design variables. An optimization problem is formulated using a set of generic target frequencies and solved using a global optimization algorithm deemed more appropriate for expensive functions, namely, a so-called surrogate optimization scheme. Our results show that tuning of the first few bending and torsional modes is feasible in practice. Moreover, from the experimental work performed on the tuned bars emerges the musical significance of the tuned torsional modes.

Subsequent to the submission of the present paper, the work of Beaton and Scavone[15] was published, similarly dealing with the concomitant tuning of bending and non-bending modes, albeit using an approach considerably distinct from ours. Their undercut model is defined by a set of points on a 2-D grid, representing the height of the undercut, which varies along the length and width of the bar. Using a gradient-based (local) optimization algorithm and appropriate initial conditions, they are able to find several interesting (and complex) undercut geometries with concomitant tuning of bending and non-bending modes. Even though experimental validation was not presented, their work shows interesting results, underlining the potential of pursuing a 3-D approach to the tuning of musical bars.

In Sec. II, we discuss the categorization and identification of the various modal families found in slender vibrating beams and underline some pertinent questions regarding the relevance of torsional modes to sound quality and timbre. Section III describes the used undercut model, the calculation of natural frequencies from a 3-D FE model, and the objective function employed in the formulation of the global optimization problem. In Sec. IV, a global optimization

https://doi.org/10.1121/10.0006573

algorithm is presented as well as some practical strategies aimed to minimize computational time. Section V presents some illustrative results to highlight the potential of the proposed approach and discuss some practical design considerations. In Sec. VI, an experimental investigation is described to validate our modeling results and clarify some aspects regarding the importance of non-bending modes to the sound quality of the tuned bars.

II. BAR MODES

A. Mode shape classification and identification

In mallet percussion instruments, bars are generally held by soft and/or elastic strings, positioned at the two nodal lines of the fundamental vertical-bending mode. Nevertheless, modeling is typically pursued using free-free boundary conditions, as the compliant supports tend to have a negligible effect on modal frequencies.[16]

In a 3-D beam with uniform cross section and free-edge boundary conditions, the various families of vibrating mode shapes are clearly distinguishable: bending modes (three directions), axial/compression modes (three directions), and torsional modes. Figure 2 illustrates the families of modes typically found in the audible frequency range. Studies in the field of mallet percussion instruments generally refer to vertical-bending and torsional modes as being the most relevant to timbre and sound quality. This hypothesis is likely based on the arguments that lateral-bending and axial modes (1) will barely be excited from a generally vertical impact of a mallet and (2) will not be very efficient acoustic radiators, since their main directions of vibration are along the slender parts of the beam. In Sec. VI D, however, some of these claims are confronted by our experimental results.

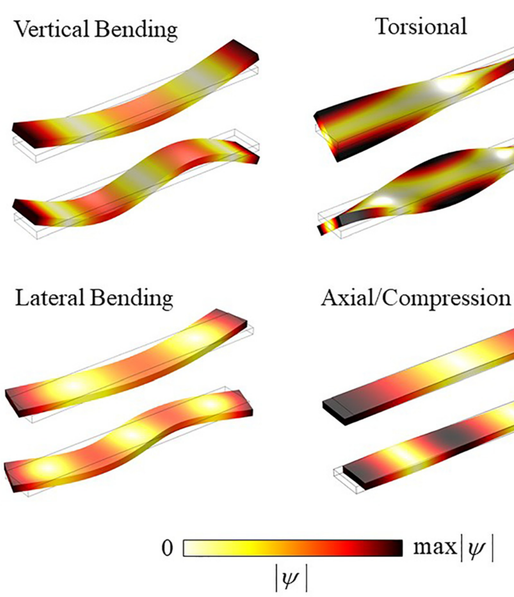

FIG. 2. (Color online) The first two mode shapes ψ of each "family" in a slender 3-D beam with uniform cross section and free-edge boundary conditions.

In most cases, the same mode type classification used in uniform beams can be made with regard to undercut beams. However, in some particular geometries, this formal classification might become ambiguous as some mode shapes pertain to combinations of different modal families. This typically occurs when two modes of distinct families are close in frequency. In these cases, care must be taken to correctly identify the modal family associated with each mode. Here, a set of simple and reliable criteria was used, based on the evaluation of the 3-D mode shape $\boldsymbol{\psi}(s) = [\psi_x(s),\ \psi_y(s),\ \psi_z(s)]$ at two corner points s_1 and s_2, as illustrated in Fig. 3.

The modal family identification process goes as follows:

- Lateral-bending modes and axial modes are identified directly if any of the conditions are true: $\max(|\boldsymbol{\psi}(s_1)|) = |\psi_y(s_1)|$ (lateral) or $\max(|\boldsymbol{\psi}(s_1)|) = |\psi_x(s_1)|$ (axial).
- If none of the above are true, i.e., $\max(|\boldsymbol{\psi}(s_1)|) = |\psi_z(s_1)|$, the mode can be either vertical-bending or torsional. Then the signs of $\psi_z(s_1)$ and $\psi_z(s_2)$ provide a disambiguation, i.e., if $\mathrm{sign}(\psi_z(s_1)) = \mathrm{sign}(\psi_z(s_2))$, the mode is vertical-bending; otherwise, if $\mathrm{sign}(\psi_z(s_1)) \neq \mathrm{sign}(\psi_z(s_2))$, the mode is torsional.

So far, the authors have seen no mis-identification using these criteria, mainly because, even in the troubling scenario of coupled modal families, there tends to be a clear domination of one modal family over the other.

B. Discussion on torsional modes

The influence of torsional modes on the bar's sound is recognized by both scientists and manufacturers.[4,11,16,17] The most demanding tuning scenarios are associated with the low register of the instrument, where, generally, the first two torsional modes appear in the frequency range of interest (typically <3 kHz). These are known to have a disturbing effect, particularly when their frequencies are close to those of the tuned bending modes, creating beating or dissonant effects. However, the lack of psychoacoustic studies on their subjective perception makes it difficult to assess their concrete influence on timbre and sound quality as well as their relative importance compared to bending modes. Moreover, it is difficult to assess which torsional modes contribute more to the beam's sound. Intuitively, their relative importance could be attributed to frequency, on the argument that lower frequency modes will generally have longer

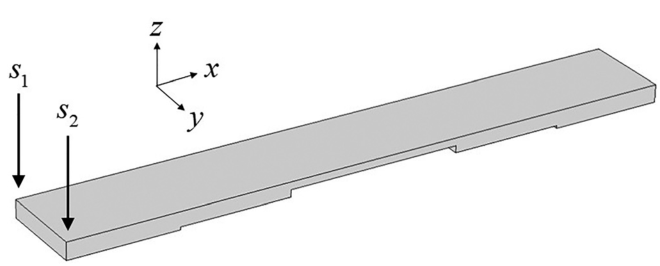

FIG. 3. Illustration of the two corner points where the mode shape is evaluated.

decay times and hence be more perceptible. However, Bork[4] reports on the adverse effect of untuned torsional modes and presents measured results that indicate the second torsional mode as the most audible amongst the non-vertical-bending modes. Naturally, this is dependent on the striking location. Nevertheless, convincing arguments can be made to classify the second torsional mode as the main contributor to timbre, aside from vertical-bending modes. First, bars are generally held by strings positioned precisely at the nodal lines of the first bending mode such that it is minimally affected by the supports. It is likely that the string supports constrict the movement of the first torsional mode (which generally has reasonable modal amplitudes at those points), adding damping and significantly lowering decay times. On this point, Oliveira and Debut[18] recently published results of an experimental modal analysis of a *timbila* (African xylophone), which emphasize the significantly larger damping ratios found for the first torsional modes. On the other hand, second torsional modes very often have nodal lines near those defined by the first vertical-bending mode, as illustrated in Fig. 4. Although dependent on the undercut geometry, this scenario is by far the most common. Second, although percussionists do strike bars at various locations, they more often tend to strike the bar vertically near its center, where the first torsional mode has a cross of nodal lines (Fig. 4). On the other hand, the second torsional mode typically has large modal displacements near the central edges of the bar. Hence, one might conjecture that, under the most common playing circumstances, the first torsional mode is typically less excited than the second. Nevertheless, these ideas have not been explored in the literature. In the experimental part of this work, the authors aim to provide an initial impulse to the discussion of some of these questions.

III. MODEL DESCRIPTION

A. Undercut model

As discussed previously, one of the key aspects to improve convergence rates during the optimization process is to have the least amount of design variables, to generate the most compact parameter space allowable. In general, if the error function describes frequency deviations only, an "ideal" global optimum (with error arbitrarily close to zero) can only be found if the number of design variables is equal to or larger than the number of defined targets.[9] In recent work,[19] the authors have used a discontinuous undercut shape with the aim of alleviating the manufacturing complexity. This profile definition provides, in a compact parameter space, a wide diversity of possible geometries.

Since manufacturers generally have the need to constrain the beam's reference dimensions, to align string supports among other practical considerations, a uniform beam with a given length L, width b, and height h_0 is considered. Its cross section is then subjected to a symmetric undercut consisting of a series of rectangular cuts, as illustrated in Fig. 5.

The undercut is then defined by $2N$ degrees of freedom (where N is the number of cuts), namely, their lengths λ_n and the associated local heights h_n. This choice of geometry leads to a set of upper and lower bounds for each variable,

$$\begin{cases} 0 \le \lambda_N \le \lambda_{N-1} \le \cdots \le \lambda_2 \le \lambda_1 \le L/2 \\ h_{\min} \le h_n \le h_0 \end{cases} \quad \text{for } n = 1, 2, 3 \ldots N, \tag{1}$$

where $h_{\min}$ is a feasible minimum thickness defined *a priori*, motivated by the need for structural integrity. Then the discontinuous cross-sectional profile $H(x)$ is given by

$$H(x) = \begin{cases} h_n & \text{if} \quad \lambda_{n+1} < x \le \lambda_n \quad \text{for } n = 1, 2, \ldots N \\ h_0 & \text{otherwise,} \end{cases} \tag{2}$$

with $\lambda_{N+1} = 0$.

This choice of undercut geometry is surely an attractive choice for metallic bars, which present relatively homogeneous material properties. However, discontinuous cuts applied to wooden bars would likely yield additional difficulties due to their naturally variable material properties. Deep localized cuts, for example, could risk the wood splitting along the grain as well as exacerbating the sensitivity of modal frequencies to small changes in geometry. Nevertheless, the present model is taken as an illustrative example of the proposed tuning procedure. The authors underline that, for wooden bars, an undercut model with

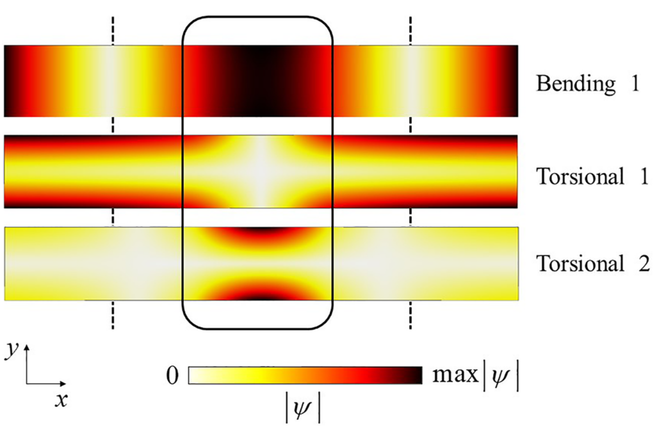

FIG. 4. (Color online) Top view of the first vertical-bending mode and the first two torsional modes of a bar with a typical tuning ratio 1:4:10, whose undercut is illustrated in Fig. 3. Support positions and typical striking area are indicated by the dashed lines and rectangle, respectively.

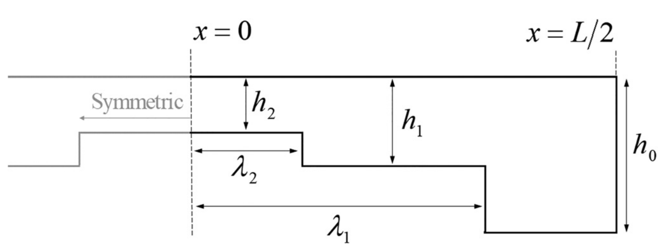

FIG. 5. Schematic description of the discontinuous undercut with $N = 2$.

https://doi.org/10.1121/10.0006573

smooth changes would be more appropriate, as found for example in the work of Henrique and Antunes.[10]

B. 3-D FE model

Calculation of modal frequencies for undercut beams was done here via 3-D FE modeling, using MATLAB's Partial Differential Equation Toolbox.[20] For simplicity, an isotropic material was considered. This assumption is reasonable when referring to metal bars, such as those found in vibraphones (aluminum). However, wooden bars in marimbas and xylophones are known to have anisotropic behavior that, in general, should not be neglected.[21] Nevertheless, the incorporation of elasticity matrices associated with particular orthotropic/anisotropic materials could easily be implemented. Moreover, damping is assumed to be weak, such that it has a negligible impact on the natural frequencies of the associated conservative beam.[21]

The meshing was composed of unstructured ten-node tetrahedral quadratic elements, eventually refined at different stages of the optimization process, as will be discussed later. Figure 6 shows two typical meshes with low and high rendering. Assuming free boundary conditions (homogeneous Neuman condition) at all nodes in exterior surfaces, the governing equations of three-dimensional elasticity, derived from Hooke's law (Navier's equations),[22] eventually lead to an eigenproblem in the typical form,

$$\left([\mathbf{K}] - \boldsymbol{\omega}_n^2[\mathbf{M}]\right)\boldsymbol{\psi}_n = \{\mathbf{0}\}, \tag{3}$$

where $\boldsymbol{\omega}_n$ and $\boldsymbol{\psi}_n$ are the natural frequencies and mode shapes of the bar, respectively, while $[\mathbf{K}]$ and $[\mathbf{M}]$ are the stiffness and mass matrices, respectively. As a note, the meshes shown in Fig. 6 would lead to an eigenvalue problem of size approximately equal to 1×10^4 and 5×10^5 for the low and high rendered meshes, respectively.

C. Objective function

For the tuning of multiple modal frequencies to a predefined target set, the optimization problem must include the deviations between modelled and target frequencies. The simplest and most commonly used approach is to define a single error function describing all frequency deviations to be minimized. A positive definite function of this type, including both bending and torsional modes, is given by

$$\varepsilon(\boldsymbol{\lambda}, \mathbf{h}) = \frac{100}{(R+M)}\left[\sum_{m=1}^{M}\left(\frac{\bar{f}_{B,m}(\boldsymbol{\lambda}, \mathbf{h}) - f_{B,m}^*}{f_{B,m}^*}\right)^2 + \sum_{r=1}^{R}\left(\frac{\bar{f}_{T,r}(\boldsymbol{\lambda}, \mathbf{h}) - f_{T,r}^*}{f_{T,r}^*}\right)^2\right], \tag{4}$$

where $f_{B,m}^*$ and $\bar{f}_{B,m}(\boldsymbol{\lambda}, \mathbf{h})$ are the target and modelled frequencies of the bending modes, respectively; $f_{T,r}^*$ and $\bar{f}_{T,r}(\boldsymbol{\lambda}, \mathbf{h})$ are the analogous for torsional modes; and M and R are the number of targets in each case.

Alternatively, one may want to tune torsional modes without specifying a particular frequency for each mode to alleviate the often-strict objective function. A more flexible criterion would be to define a set of possible target frequencies $\mathbf{f}_2^*$ in which the torsional modes *can* lie. In this case, the target frequencies $f_{2,r}^*$ would be those, contained in $\mathbf{f}_2^*$, that satisfy the condition $\min|\bar{f}_{2,r} - \mathbf{f}_2{}^*|$, i.e., those nearest to the modelled frequencies.

D. Optional additional penalties

In many cases, there are multiple feasible solutions for a specific target set. Hence, one may want to optimize other aspects of the beam geometry in addition to tuning its modal frequencies.

As an example, we illustrate here a previously studied approach[19] to minimize the amount of material extracted, which may be an important consideration from the point of view of manufacturing. The percentage of volume of extracted material, with respect to the original bar volume (uniform beam), is given by

$$V(\boldsymbol{\lambda}, \mathbf{h}) = 100 \cdot \frac{2}{h_0 L}\int_0^{L/2}(h_0 - H(x))dx. \tag{5}$$

Then a combined objective function might be formulated as follows:

$$\mathrm{E}(\boldsymbol{\lambda}, \mathbf{h}) = (1-\alpha)\varepsilon + \alpha V, \tag{6}$$

where $0 \le \alpha \le 1$ is a weighting factor on the volumetric penalty, i.e., if $\alpha = 0$, the volumetric constraint is not accounted for; if $\alpha = 1$, only the volumetric condition is accounted for. In our recent work,[19] the interested reader can find a report on the impacts of using such penalty as well as appropriate values for the weighting factor α. Here, the function V may be replaced by any other to satisfy a particular characteristic of the beam geometry. For example, a penalty for tuning "robustness" could also be considered. That is, by minimizing a function of the type $|\partial\varepsilon/\partial\mathbf{x}|$, where $\mathbf{x} = \{\boldsymbol{\lambda}, \mathbf{h}\}$, the algorithm will aim to generate geometries that are less sensitive to slight machining/modeling errors. Another possibility would be to consider some characteristic

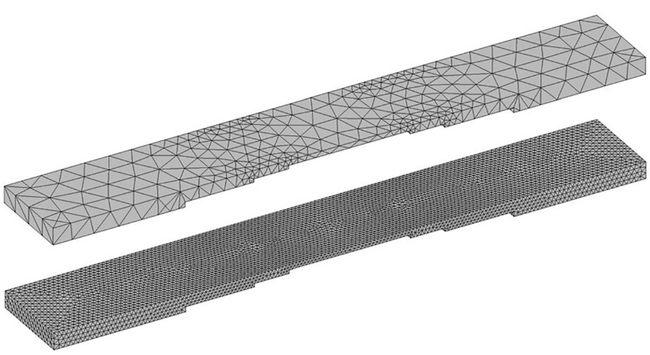

FIG. 6. Example of the tetrahedral meshing with different rendering.

of the beam's mode shapes that would improve acoustic radiation, support positioning, etc.

IV. OPTIMIZATION METHODS

As mentioned before, the use of 3-D FE models to predict the modal frequencies of the beams demands additional attention to the optimization methods to use. Contrary to simpler 1-D FE models, the eigenvalue problem associated with a 3-D meshed beam generally takes between 1 and 10 s to solve, depending on the mesh used. This underlines the importance of choosing an appropriate optimization scheme that ideally minimizes the number of function evaluations necessary to reach a viable solution.

Moreover, since our objective function is known to have many sub-optimal local minima,[9,10,19] the use of efficient gradient-based optimization schemes is not adequate, and global optimization schemes are, in general, more appropriate. With that said, at a later stage in the optimization process, when a feasible region of solutions (valley) has been identified by a global optimization algorithm, gradient-based optimization methods can be useful as they will find the nearest local minima using a minimal amount of function evaluations. In this work, we have often used this approach as a final stage in the optimization process, using MATLAB's fmincon() function.[23]

There are a large variety of global optimization methods available: evolutionary algorithms, simulated annealing, particle-swarm, etc. However, the most commonly used methods frequently rely on a large number of function evaluations.[24] Consequently, attention is devoted to global optimization schemes deemed more appropriate for expensive functions, namely, those using radial basis functions, commonly called surrogate optimization schemes.

A. Surrogate optimization

The essence of surrogate optimization algorithms relies on the creation of a (surrogate) function that approximates the objective function based on an interpolation scheme using only a few function evaluations. The surrogate function serves then as a guide to generate appropriate candidate solutions. Once the new candidate solutions are evaluated, a new interpolation is calculated, updating the surrogate function and generating new candidate solutions recursively. Hence, a surrogate optimization algorithm alternates between two stages: (1) construction of a surrogate model and (2) the search for a minimum using the surrogate function as guide. Here, only a brief summary of the procedures involved in a surrogate optimization algorithm is presented. For a detailed description, see, for example, Refs. 25–27.

In the initial stage, the algorithm evaluates the expensive objective function $f(x)$ at m random points $[\{x_1\}, \{x_2\} \dots \{x_m\}]$ within the bounded parameter space $\mathbb{R}^N$, resulting in the data points $\{f(x_1), f(x_2), \dots, f(x_m)\}$. Then a surrogate function $s(x)$ that interpolates the various data pairs (x_m, f_m) can be formulated in the form

$$s(x) = \sum_{i=1}^{m} \beta_i\, \phi(\|x - x_i\|) \qquad x \in \mathbb{R}^N, \tag{7}$$

where $\|\cdot\|$ is the Euclidian norm; $\beta_i \in \mathbb{R}$ for $i = 1, 2, \dots m$; and $\phi(r)$ is the radial basis function to be chosen [e.g., $\phi(r) = r^3$]. Ultimately, this interpolation procedure reduces to solving an $m - \text{by} - m$ linear system of equations for the coefficients β_i.

Once a surrogate function $s(x)$ is calculated, the algorithm searches for new candidate solutions. New candidate solutions are chosen on two criteria: (1) minimizing the surrogate function and (2) being located at a reasonable distance from already evaluated points. This enables a balance between minimizing the objective function and providing a robust search in the parameter space. In summary, the algorithm generates thousands of sample points (since the evaluation of the surrogate is computationally cheap), and a new candidate solution is chosen such that it minimizes a certain *merit* function, given for example by

$$M(x) = \alpha S(x) + (1 - \alpha) D(x), \tag{8}$$

where α is a weighting factor $0 < \alpha < 1$. $S(x)$ and $D(x)$ are the normalized surrogate function and normalized distance to previously evaluated points, given by

$$S(x) = \frac{s(x) - s_{\min}}{s_{\max} - s_{\min}}; \quad D(x) = \frac{d_{\max} - d(x)}{d_{\max} - d_{\min}}, \tag{9}$$

where $s_{\min}$ and $s_{\max}$ are the minimum and maximum surrogate values amongst the sampled points, while $d_{\min}$ and $d_{\max}$ are the minimum and maximum distance between the sampled point and an evaluated point. Subsequently, the value of the "expensive" objective function for the new candidate solution is calculated, and the procedure continues repeatedly.

Detailed descriptions of the algorithm can be found in Refs. 25–28 as well as proof of global convergence in a bounded domain and various numerical studies demonstrating the advantages of the algorithm over other global optimization schemes. In this work, we have used a version contained in MATLAB's Global Optimization Toolbox, surrogateopt().[29]

B. Approaches to further reduce computational time

The optimization procedure described above works as a stand-alone method and, in general, is able to find feasible solutions in reasonable amounts of computational time. However, additional strategies can further reduce the amount of computational time necessary, particularly when the defined tuning targets are demanding.

First, one may wish to initially optimize the tuning of the bending modes only, using an appropriate 1-D model for low computational cost. Once a feasible solution $\{x_0\}$ is found, it can be inserted as an "initial search point" in the surrogate model, i.e., forcing the initial set of random points $\mathbf{x}_m$ to be $\mathbf{x}_m = [\{x_0\}, \{x_1\}, \{x_2\} \dots \{x_m\}]$, now using an objective function with both bending and torsional modes. Moreover, the torsional modes of the initial solution $\{x_0\}$ can be monitored, and one may want to choose the tuning frequencies of the torsional modes such that they are closer to those found in the initial solution.

Second, since the mesh rendering will affect significantly the computational time associated with a function evaluation, it can be advantageous to start the optimization process on a low rendered mesh at first and, at a later stage, refine the mesh for increased accuracy in the final solution. In this case, knowing that poorly rendered meshes add some "artificial" stiffness to the model,[30] the target frequencies at the low-rendering stage should be slightly increased (typically by 1%–2%).

Finally, one could also exploit the symmetry of the bars to build models with a lower number of degrees of freedom. For example, if we consider only half the bar domain (e.g., positive x space), we could add symmetric/anti-symmetric conditions at the boundary $x = 0$. However, because mode shapes have different symmetry properties, the reduced model would have to be solved twice, with both symmetric and anti-symmetric boundary conditions, to capture all modes. Similarly, one could account for the two-way symmetry (i.e., planes $x = 0$ and $y = 0$) and reduce the domain to a quarter of the original size. In this case, the reduced model would have to be solved four times (all combinations of symmetry and anti-symmetry) for all modes to be calculated. Even though we have not used this approach in the present work, for programming simplicity, some preliminary studies indicate that solving the original system once yields larger computational times than solving the reduced system multiple times.

V. ILLUSTRATIVE EXAMPLES

A. Tuning a single torsional mode

To illustrate the potential of the proposed approach, we now consider an aluminum bar with reference length $L = 350$ mm, width $b = 50$ mm, and height $h_0 = 10$ mm. The chosen dimensions approximate those found in F3 bars ($\sim$175 Hz) of a typical commercial vibraphone. The material properties are taken as Young's modulus $E = 70$ GPa, density $\rho = 2750\ \text{kg} \cdot m^{-3}$, and Poisson's ratio $\upsilon = 0.35$. As mentioned before, most vibraphone (or marimba) bars are tuned such that the first three vertical-bending modes obey the ratio $f_m/f_{m=1} = (1:4:10)$. Following this convention, we will search for solutions with the same tuning ratios for the bending modes and add another target for the first torsional mode, using an undercut with three discontinuities, $N = 3$. The fundamental target frequency was set to $f^*_{B1} = 175$ Hz. For now, we neglect the tuning of the second torsional mode. An upper limit for the size of the elements in the FE mesh was set to 4 mm, leading to meshes similar to the illustration in Fig. 6 (bottom). Optimization was performed using the surrogate algorithm, and the computation was stopped once a solution with an error below the tolerance $\varepsilon_{\text{lim}} < 10^{-3}$ was found. Figure 7 shows the resulting optimized undercut profiles for bars with a $(1:4:10)$ tuning ratio for the bending modes and the first torsional mode tuned to 4, 5, 6, 7, and 8 times the fundamental.

Figure 7 shows that the bending mode tuning ratio (1:4:10) allows for some degree of flexibility in the placement of the first torsional mode. We note that, for an increasing target frequency of the first torsional mode, the optimized undercut profiles tend to have more localized and deep cuts. These results suggest that, in this particular case, the first torsional mode would more naturally fall between 4 and 5 times the fundamental, where the undercut profiles are smoother. It must be noted that attempts were made to find solutions where the frequency of the first torsional mode is 3 and 9 times the fundamental, but without success. Although difficult to prove formally, it is likely that solutions to these targets fall outside the defined parameter space. In these cases, an increase in the number of design variables $(N > 3)$ or modifications to the reference dimensions would likely be necessary to reach these demanding targets.

B. Tuning two torsional modes

Going one step further, we now consider the same reference beam, with a tuning ratio $(1:4:10|5)$ and add a target frequency for the second torsional mode in the harmonic series. Figure 8 shows the optimized profiles with the

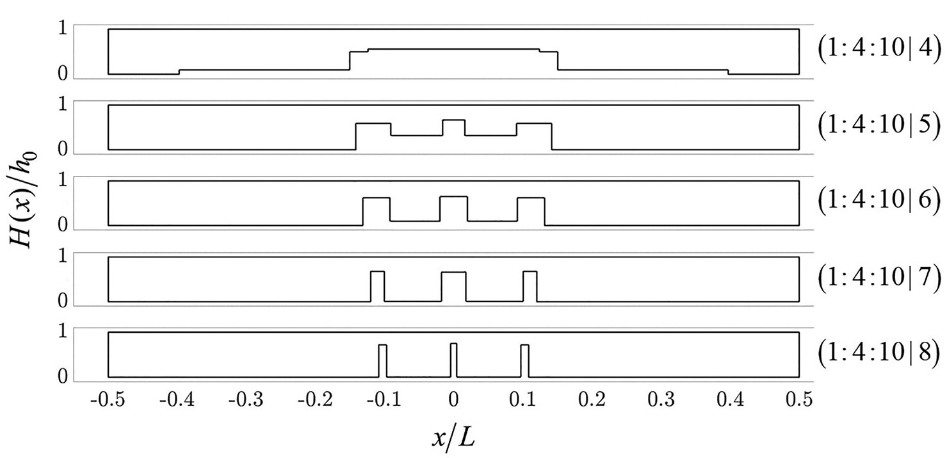

FIG. 7. Optimized undercut profiles for aluminum bars with the tuning ratio $(1:4:10|k)$, where k refers to the tuning ratio of the first torsional mode, $f_{r=1}/f_{m=1} = k$. Note that bar lengths and heights are not shown to the same scale.

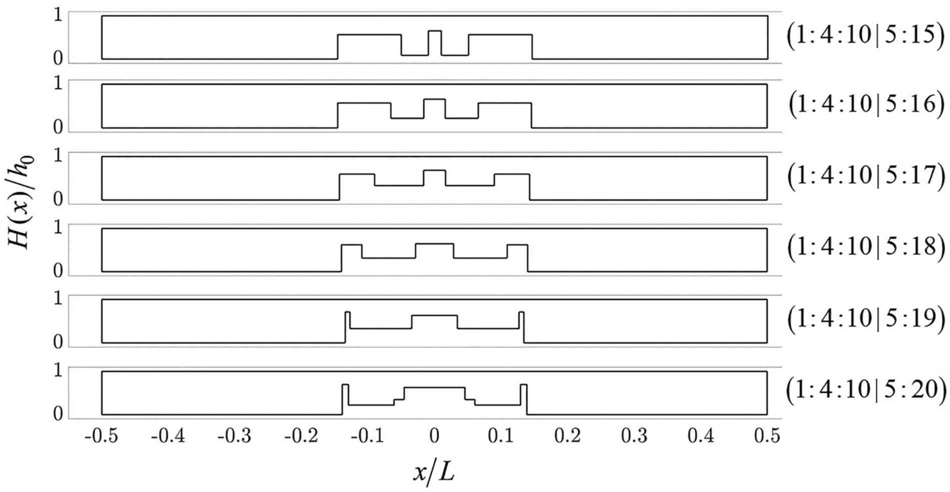

FIG. 8. Optimized undercut profiles for aluminum bars with the tuning ratio $(1:4:10|5:k)$, where k refers to the tuning ratio of the second torsional mode $f_{r=2}/f_{m=1}=k$. Note that bar lengths and heights are not shown to the same scale.

additional target for the second torsional mode at 15, 16, 17, 18, 19, and 20 times the fundamental.

While our aim is not to judge the relative merits (or otherwise) of specific tuning targets but rather illustrate the array of choices presented to manufacturers, it is useful to discuss some elementary characteristics presented by particular tunings/profiles. In regard to the examples shown in Figs. 7 and 8, we point out the following considerations, which are applicable to any generic choice of tuning targets:

(1) *Unison tuning*: In the tuning target $(1:4:10\,|\,4)$, the first torsional mode is tuned in unison with the second bending mode. This interesting choice can be aimed to reduce spectral density of the radiated sound. However, slight frequency deviations will lead to beating effects, which may be musically interesting or a nuisance, depending on the context. Hence, the tuning accuracy of these two modes is particularly demanding.
(2) *Awkward intervals*: The tuning target $(1:4:10\,|\,7)$ is rather unusual, since the seventh harmonic corresponds approximately to an interval of 2 octaves plus a minor seven (moreover out of tune in the equal tempered scale). Typical tuning ratios found in traditional instruments are generally composed of more regular musical intervals: octaves, fifths, or thirds. With that said, these choices could also lead to interesting heterodox sounds.
(3) *Profile practicality*: In many cases, the resulting optimized profile might have some impractical features, like the deep localized cuts in $(1:4:10\,|\,8)$ or $(1:4:10\,|\,5:20)$. Although perfectly viable in a musical sense, these profiles will have regions of highly localized stress and might be less robust in terms of tuning, i.e., slight modeling/manufacturing errors can induce large deviations in the natural frequencies. Moreover, potential effects in the modal damping and the practical limitations of machining need also to be considered.
(4) *Tuning frequency range*: The profiles presented in Fig. 8 consider the tuning of two torsional modes, leading to modal frequencies up to 20 times the fundamental. However, the frequency of the fourth bending mode was neglected, and indeed it falls below the frequency of the second torsional mode in all cases. Generally, it is more adequate to define an upper frequency limit and tune all modes in that range.

VI. EXPERIMENTAL VALIDATION

Several experimental investigations were performed to validate the modeling results from our 3-D FE model. In this section, we describe the experimental modal identification procedure, the estimation of material properties, and the validation of modelled natural frequencies; present a short discussion on modal damping; and finally present some spectrograms of recorded sound from the optimized bars. Throughout this section, we will refer to each mode with the notation B, T, and L, for the vertical-bending, torsional, and lateral-bending modes, respectively, i.e., bending modes are B1, B2, B3, etc., torsional modes are T1, T2, T3, etc. Additionally, we will describe frequency deviations in *cents*, a unit commonly used to quantify mistuning in a musical context. This is calculated by

$$E_{cent} = 1200\log_2\left(\bar{f}/f^*\right), \tag{10}$$

where $\bar{f}$ and f^* are the measured and target frequencies, respectively. Note that 100 cents correspond to a deviation of one semi-tone in the equal tempered musical scale.

The experimental setup for measuring the modal parameters of the experimental bars is illustrated in Fig. 9. Bars were held by two thin rubber strings placed at the nodal lines of the first vertical-bending mode (nodal lines estimated using FE model). The position of the supports was set to mimic conditions found in real instruments. In the assumption that the supports might add damping to certain modes, we can investigate how modal damping is affected in more realistic support conditions. A laser velocimeter [Polytech (Hudson, MA) PDV 100] measured the vertical velocity component at one corner of the beam while a small impact hammer [PCB Piezotronics (Depew, NY) PCB-084A17] was used to excite the beam vertically. The transfer function mobility $H_p(f)$ was measured at 16 points,

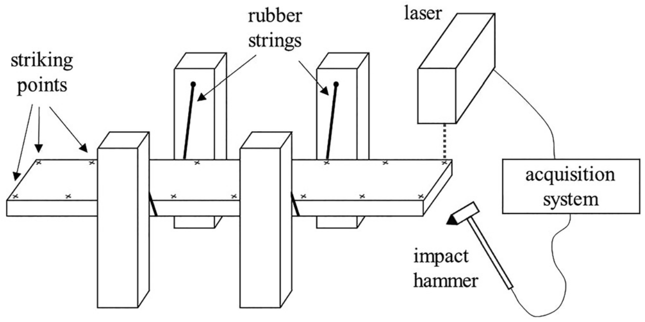

FIG. 9. Diagram of the experimental setup.

$p = 1, 2\ldots16$, along the edge of the beam. Identification of modal frequencies, modal damping, and mode shapes was done using an eigensystem realization algorithm.[31,32] This algorithm is robust and, more importantly for us, is able to identify modes with closely spaced frequencies. In our case, this is required as torsional and bending modes may be very close in frequency.

A. Measurement of material properties

To avoid difficulties associated with anisotropic behavior, we considered aluminum bars to validate our modeling results. The first step was to measure the material properties of the aluminum used for the experimental bars. Assuming the aluminum is isotropic, we need to measure the density ρ, the Young's modulus E, and the Poisson's ratio υ. To this end, we examined one aluminum bar of uniform cross section and reference dimensions: $L = 350$ mm, $b = 50$ mm, and $h_0 = 10$ mm. The density was calculated directly from weight measurements and geometric considerations, leading to $\rho \approx 2748\ \mathrm{kg} \cdot m^{-3}$.

The Poisson's ratio describes the ratio of transverse and axial strain in a particular material and will affect both bending and torsional modes. However, in slender beams, the Poisson's ratio has only a minor effect on the frequencies of the bending modes while having a crucial effect on torsional modes. Therefore, the Young's modulus was estimated based on a least-square fitting between the modelled and measured frequencies of the vertical-bending modes only, fixing the Poisson's ratio to $\upsilon = 0.3$.

To ensure discretization errors could be considered negligible, the meshing was refined by setting the maximum element size to 2.5 mm, leading to a system with approximately 8×10^5 degrees of freedom. To illustrate the order of magnitude of meshing errors, a similar system with maximum element size 3 mm (450 000 degrees of freedom) yielded frequency deviations of approximately 0.002 cents for the lowest bending mode and 0.05 cents for the sixth vertical-bending mode, compared to the results using the refined mesh.

The best-fit for the frequencies of vertical-bending modes resulted in $E = 69.8$ GPa. Subsequently, the Poisson's ratio was estimated in a similar manner, by fixing the Young's modulus ($E = 69.8$ GPa) and fitting the frequencies of the torsional modes, leading to $\upsilon = 0.28$. The deviation between the measured and modelled frequencies is shown in Fig. 10. This procedure allows for a rough estimation of the Young's modulus and Poisson's ratio in a simple manner. Note, however, that other methods exist to obtain more accurate estimations of these material properties from modal identification.[33–35]

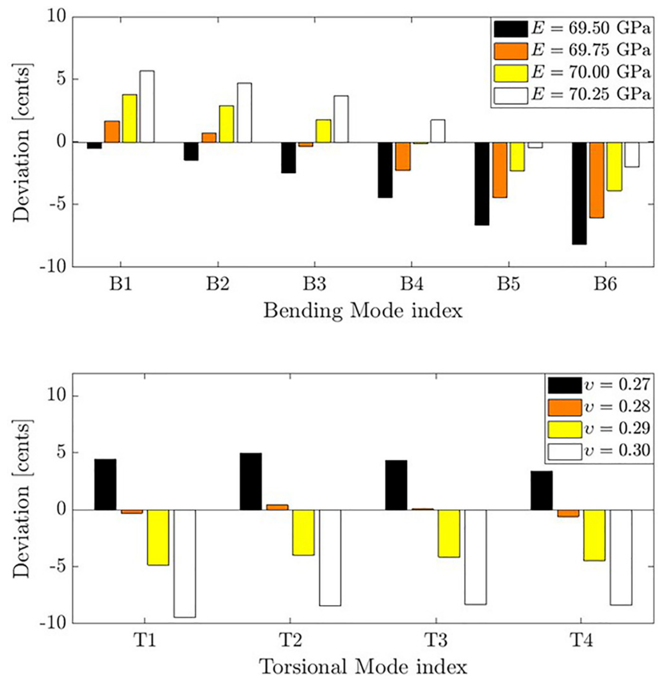

FIG. 10. (Color online) Deviation (in cents) between the measured and modelled modes. Bending modes are shown on the top for various values of E(with $\upsilon = 0.3$), while torsional modes are shown on the bottom for various values of υ (with $E = 69.8$ GPa).

B. Validation of modelled frequencies

Given an estimation of the material properties of the aluminum, several optimized profiles were calculated for tuning both bending and torsional modes. We defined the fundamental frequency to be $f_1 = 175$ Hz (near the musical note F3) and decided to tune bar modes up to a frequency of 3500 Hz. In the three examples that follow, we have fixed the tuning of the first four vertical-bending modes to (1:4:10:16), the tuning ratio of the first three (1:4:10) being the most commonly found in marimbas and vibraphones. For the first two torsional modes, we have defined the targets ratios (4:16), (5:20), and (6:18). The resulting optimized profiles were then manufactured in a CNC milling machine, capable of a cutting accuracy of approximately ± 0.05 mm. The manufactured bars are shown in Fig. 11.

The first design (1:4:10:16 | 4:16) presents monotonic profile changes, with less abrupt discontinuities. More notably, the two torsional modes T1 and T2 are tuned to the same frequency of the two bending modes B2 and B4. This would lead to a reduction of the number of frequencies radiated by the bar but may also suffer from unwanted beating effects if the modes are not accurately tuned. The other two designs (1:4:10:16 | 6:18) and (1:4:10:16 | 5:20) have more

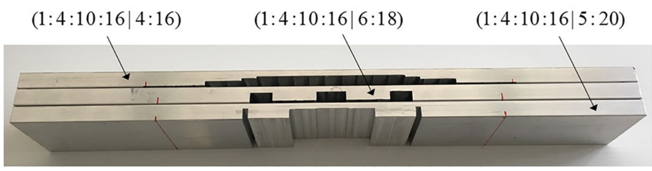

FIG. 11. (Color online) The three manufactured bars and their respective target tuning ratios.

abrupt profile discontinuities, particularly the latter, with two very deep localized cuts. This will allow us to assess the viability of these types of designs in which stresses are more localized.

A comprehensive overview of the measured (mobility) transfer functions can be illustrated by an average power spectrum (APS), given by

$$\mathrm{APS}\,(f) = \frac{1}{P}\sum_{p=1}^{P} |H_p(f)|^2, \tag{11}$$

where $|H_p(f)|$ are the magnitudes of the measured transfer functions and here $P = 16$. The APS for each bar is shown in Fig. 12, as a function of the frequency normalized by the fundamental frequency f_1. Note how the spectrum of the first bar (1:4:10:16 | 4:16) is less dense because two pairs of modes are tuned to the same frequency. In this case, up to 3500 Hz ($f/f_1 \approx 20$), the bar will radiate energy in only four frequencies compared to the expected six found in the other two bars.

To illustrate the tuning accuracy of the manufactured bars and validate our modeling results, we show in Table I the deviation between the measured and target tuning ratios.

Overall, the modal frequencies of the manufactured bars showed relatively small deviations compared to the expected target, demonstrating the benefit of using 3-D FE models. In the first two designs (1:4:10:16 | 4:16) and (1:4:10:16 | 6:18), all modal frequencies had absolute deviations smaller than 16 cents, with an average deviation of 9.0 and 10.9, respectively. In the third design (1:4:10:16 | 5:20), we notice larger deviations, up to 46.7 cents with an average of 32.9 cents. It is likely that these larger errors are associated with the abrupt profile discontinuity of this design. The presence of deep localized cuts might result in a design that is less robust to machining errors, as small inaccuracies in manufacturing will lead to larger changes in the modal frequencies.

Aside from the deviations in the tuning ratios, it is worth noting that measured modal frequencies were somewhat lower than expected overall. For the (1:4:10:16 | 4:16), (1:4:10:16 | 6:18), and (1:4:10:16 | 5:20) designs, we had fundamental frequencies of 172.4, 173.8, and 171.2 Hz, respectively, compared to the expected 175 Hz. These deviations correspond to errors of −26, –12, and −38 cents, respectively. However, since all modal frequencies were affected in a similar manner (i.e., tuning ratios remained reasonably accurate), the most likely explanation is that the

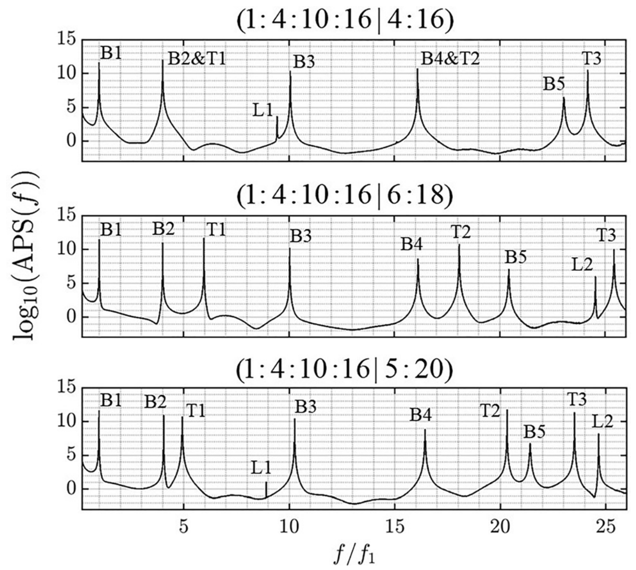

FIG. 12. APS of the three manufactured bars. The notation B, T, and L is used to identify the vertical-bending, torsional, and lateral-modes, respectively.

Young's modulus of bars decreased in the milling process, as is known to happen when aluminum alloys are exposed to high temperatures.[36–38] During milling, temperatures near the tool-material interface are known to reach between 250 °C and 650 °C, depending on a variety of parameters like cutting speed and depth, tool material and sharpness, lubrication, etc.[39] If that is the case, results suggest that, using pre-treated materials or a less invasive machining procedure to prevent high temperatures (slower cutting velocities, extensive use of cooling liquid, etc.), tuning accuracies could possibly be improved even further.

Here, it is worth noting that mistuning perception thresholds are estimated around ±2–3 cents in musically trained listeners and ±12–16 cents for lay listeners, concerning pure and complex tones.[40,41] However, in the context of mistuned harmonic partials, perception thresholds are typically larger, estimated around 5–40 cents, increasing in proportion to harmonic order.[42,43] With this in mind, the presented results demonstrate the practical potential of the proposed design-optimization procedure. It shows that bars with demanding tuning targets can be easily manufactured with small/negligible tuning errors. Aside from allowing the tuning of multiple modes to a generic set of predefined frequency targets, it could potentially enable a bypass on the human intervention altogether, as is typical in today's manufacturing contexts.

TABLE I. Deviation between measured to target tuning ratios in cents, E_{cent}.

	B1	B2	B3	B4	T1	T2	avg($\lvert E_{cent}\rvert$)
(4:16)	Ø	3.8	11.7	13.8	4.4	11.4	9.0
(6:18)	Ø	6.7	9.1	16.0	–7.1	10.1	10.9
(5:20)	Ø	22.5	43.7	46.7	–24.2	27.5	32.9

https://doi.org/10.1121/10.0006573

C. Discussion on modal damping

In a typical mallet instrument setup, energy dissipation from a vibrating bar can presumably be caused by (1) internal losses, associated with material viscosity effects, (2) acoustic radiation, and (3) coupling effects due to support conditions. The modal damping ratios ζ were identified experimentally to investigate our previous hypothesis concerning the relative importance of the torsional modes to timbre.[44] Figure 13 shows the measured modal damping ratios for the first five bending modes and first three torsional modes of the experimental bars.

Overall, we notice that all modal damping ratios remained within the same order of magnitude (approximately 0.01%–0.1%), in all three bars. Noticeably, the second torsional mode was consistently the lowest damped mode in each bar, never exceeding 0.01%. This is in agreement with our previous conjecture. The damping of the first torsional mode is not particularly large, indicating that this mode might indeed be noticeable and important to timbre. However, it must be said that support conditions here (very thin elastic string) differ from those found in real instruments, which may be more invasive, using thicker and/or less elastic strings. Nevertheless, these preliminary results suggest that this mode should not be neglected in the design process.

D. Timbral features

While it is not the aim of this work to assess the sound quality of bars with tuned torsional modes, we feel it is useful for the reader to hear what these bars could sound like. Hence, we provide with the article sound recordings of the experimental bars (Mm. 1, Mm. 2, Mm. 3, Mm. 4). The bars were held by the same support conditions described earlier. Additionally, cylindrical (resonator) tubes were placed beneath the bar to simulate conditions found in real instruments. The resonator tubes were made of PVC and had a diameter of 50 mm (same as the width of the bars), and their length was varied such that their fundamental acoustic mode matched the frequency of the first bending bar mode, as is commonly seen in vibraphones and marimbas. A condenser microphone was placed 50 cm above the bar in an off-centered location. The bars were excited vertically by the stroke of a typical vibraphone mallet. The striking point was at an off-centered location, appropriate for the excitation of both bending and torsional modes. Figure 14 shows the spectrogram of the sounds recorded for the three experimental bars described above as well as that of a bar without tuning of the torsional modes.

Mm. 1. Sound recording of the optimized bar with tuning ratio (1:4:10:16 | 4:16). This is a file of type "wav" (2.0 MB).

Mm. 2. Sound recording of the optimized bar with tuning ratio (1:4:10:16 | 6:18). This is a file of type "wav" (1.9 MB).

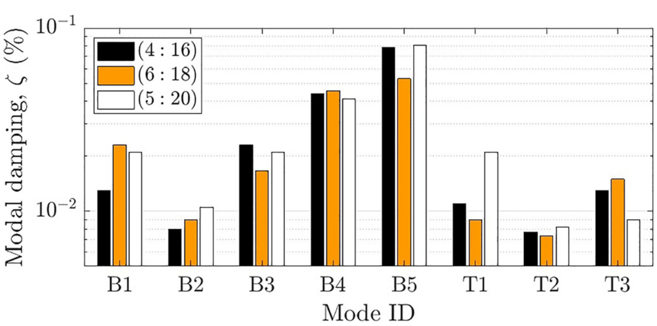

FIG. 13. (Color online) Modal damping ratios $\zeta(\%)$ of the experimental bars.

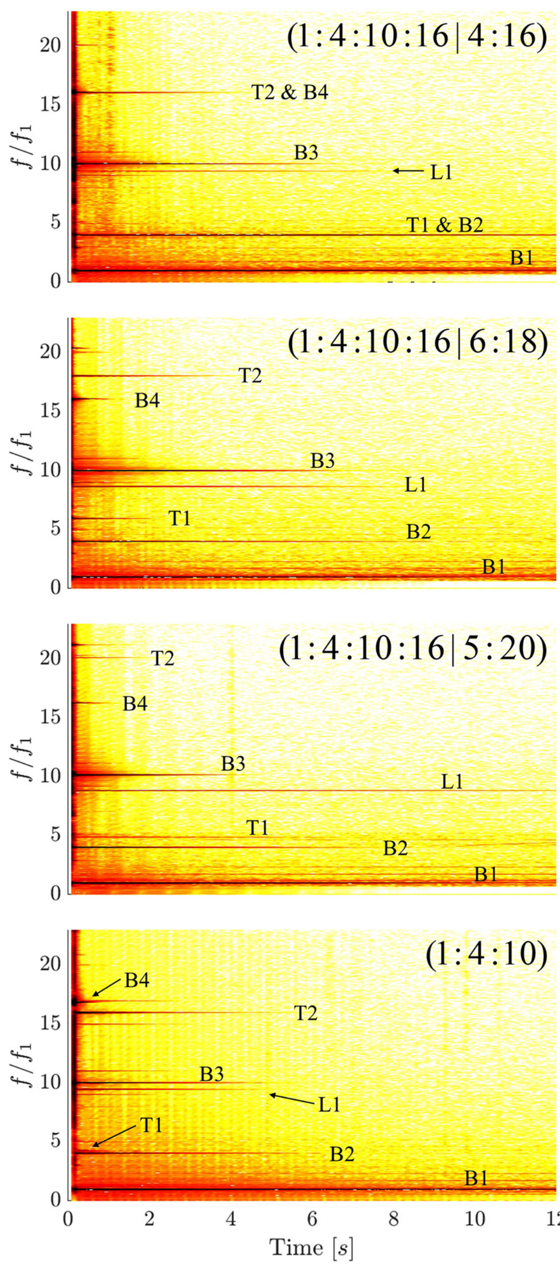

FIG. 14. (Color online) Spectrograms of the radiated pressure from the three tuned experimental bars and from a bar without torsional tuning. As before, the modes are identified by the notation B, T, and L for the bending, torsional, and lateral modes, respectively.

https://doi.org/10.1121/10.0006573

Mm. 3. Sound recording of the optimized bar with tuning ratio (1:4:10:16 | 5:20). This is a file of type "wav" (2.0 MB).

Mm. 4. Sound recording of the bar without tuning of torsional modes. This is a file of type "wav" (1.8 MB).

From the spectrograms of recorded sound, we notice that the first three vertical-bending modes (B1, B2, and B3) are the most prominent, as expected. The higher order vertical-bending modes (B4 and B5) appear to radiate efficiently but generally have very low decay times, meaning that their influence on timbre will likely be associated with the "striking sound." Both torsional modes (T1 and T2) are clearly seen, with the second torsional mode often having a longer decay time, in accordance with the measured low modal damping discussed in Sec. VI C.

It is interesting that in some cases, we see frequency components that do not belong to any particular mode. These typically appear in pairs, equidistant to a particular mode [note, e.g., modes B3 and T2 in (1:4:10) or B3 in (1:4:10:16 | 4:16)], seemingly following a relation of the type $f_X \pm n f_{B1}$, where f_X is the frequency of either a bending or torsional mode. These effects are typical signatures of mode coupling resulting from geometric nonlinearities.[17,45]

Finally, we see a very significant appearance of the first lateral-bending mode in all recordings. Unexpectedly, this mode appears to radiate reasonably, despite its main direction of vibration being along the slender part of the beam. As mentioned earlier, because undercut bars do not have in-plane symmetry, lateral modes will also have small vertical components. Examining the first seconds after the transient, we see that this mode does not radiate as strongly as vertical-bending and torsional modes. However, it appears to have extremely low damping, and its presence becomes significant once the other modes decay. This effect is more prominently seen/heard in the recording of the (1:4:10:16 | 5:20) bar. The exceptionally low damping of this mode might be attributed to several causes. First is the fact that the nodes of this mode are located near the attachment points of the bar, preventing localized energy dissipation by the supports. Second is the longer vibratory motion of this mode due to its inefficient acoustic radiation. With that said, it is likely that this mode is less relevant in wooden bars, where damping is much larger. Even though we have not accounted for this mode in the optimization, this emphasizes the central aim of our work: the importance of tuning non-vertical-bending modes. We anticipate that, equipped with the proposed approach, future work will deal with the concomitant tuning of vertical-bending, torsional, and lateral-bending modes.

VII. CONCLUSIONS

In this paper, the viability of designing percussive bars with both bending and torsional modes precisely tuned was demonstrated. The employed undercut model consists of a series of symmetric discontinuous cuts aimed to facilitate the manufacturing process as well as providing a versatile geometry using a minimal set of design variables. The use of 3-D FE models allowed the calculation of the frequencies of both bending and torsional modes and also provided an increased accuracy compared to the commonly used 1-D models. A surrogate function algorithm was used to solve the global optimization problem, enabling fast computations with the naturally expensive function evaluations associated with 3-D FE models. Results from an experimental modal identification of optimized bars with demanding tuning targets positively validated our modeling results, with tuning deviations generally falling below 15 cents. Our preliminary experiments also suggested that typical support locations might contribute to a particularly low damping of the second torsional mode. In the future, we hope to assess the effect of realistic support conditions on modal damping, with the aim to evaluate the relative importance of the first and second torsional modes to timbre. Finally, spectrograms of recorded sound from the experimental bars indicate that, surprisingly, the first lateral-bending mode might have a significant influence on timbre and should, in some contexts, be included in the optimization procedure. In light of the promising viability results shown here, it is expected that tuning of the first few torsional modes will be routinely applied to high quality instruments in the near future.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the financial support given by "Fundação para a Ciência e Tecnologia" (FCT–Portugal) through Ph.D. Grant No. SFRH/BD/140598/2018. Additionally, we would like to thank José Guilherme Ferreira for his diligent work in manufacturing the optimized bars.

[1]N. Fletcher and T. Rossing, *The Physics of Musical Instruments* (Springer-Verlag, New York, 1998).

[2]F. Orduña-Bustamante, "Nonuniform beams with harmonically related overtones for use in percussion instruments," J. Acoust. Soc. Am. **90**(6), 2935–2941 (1991).

[3]I. Bork, "Vibration modes and tuning of xylophone bars," in *Proceedings of the 117th Meeting of the Acoustical Society of America*, Syracuse, New York (May 22–26, 1989).

[4]I. Bork, "Practical tuning of xylophone bars and resonators," Appl. Acoust. **46**, 103–127 (1995).

[5]I. Bork, "Measuring the Acoustical Properties of Mallets," Appl. Acoust. **30**, 207–218 (1990).

[6]I. Bork and J. Meyer, "On the tonal evaluation of xylophones," Percussive Notes **23**, 48–57 (1985).

[7]I. Summers, S. Elsworth, and R. Knight, "Transverse vibrational modes of a simple undercut beam: An investigation of overtone tuning for keyed percussion instruments," Acoust. Lett. **17**, 66–70 (1993).

[8]J. Petrolito and K. A. Legge, "Optimal undercuts for the tuning of percussive beams," J. Acoust. Soc. Am. **102**(4), 2432–2436 (1997).

[9]J. Petrolito and K. A. Legge, "Designing musical structures using a constrained optimization approach," J. Acoust. Soc. Am. **117**(1), 384–390 (2005).

[10]L. Henrique and J. Antunes, "Optimal design and physical modelling of mallet percussion instruments," Acta Acust. **89**, 948–963 (2003).

[11]D. Beaton and G. Scavone, "Optimization of marimba bar geometry by 3D finite element analysis," in *Proceedings of ISMA 2019*, Detmold, Germany (September 13–17, 2019).

https://doi.org/10.1121/10.0006573

[12]J. Bretos, C. Santamaría, and J. A. Moral, "Finite element analysis and experimental measurements of natural eigenmodes and random responses of wooden bars used in musical instruments," Appl. Acoust. **56**, 141–156 (1999).
[13]D. Beaton and G. Scavone, "Measurement-based comparison of marimba bar modal behaviour," in *Proceedings of ISMA 2019*, Detmold, Germany (September 13–17, 2019).
[14]B. Kirkland and L. Moradi, "Topographical optimization of structures for use in musical instruments and other applications," in *Proceedings of the Society for Design and Process Science—21st International Conference*, Orlando, FL (December 4–6, 2016).
[15]D. Beaton and G. Scavone, "Three-dimensional tuning of idiophone bar modes via finite element analysis," J. Acoust. Soc. Am. **149**(6), 3758–3768 (2021).
[16]A. Chaigne and J. Kergomard, *Acoustics of Musical Instruments* (Springer-Verlag, New York, 2016).
[17]T. Rossing, *Science of Percussion Instruments* (World Scientific Publishing, Singapore, 2000).
[18]E. Oliveira and V. Debut, "Geometry and tuning assessment of timbilas through non-destructive reverse engineering techniques," in *Proceedings of Acústica 2020* (October 21–23, 2020).
[19]F. Soares, J. Antunes, and V. Debut, "Multi-modal tuning of vibrating bars with simplified undercuts using an evolutionary optimization algorithm," Appl. Acoust. **173**, 107704 (2021).
[20]MathWorks, "Partial differential equation toolbox," https://www.mathworks.com/products/pde.html (Last viewed September 28, 2021).
[21]I. Bork, A. Chaigne, L.-C. Trebucher, M. Kosfelder, and D. Pillot, "Comparison between modal analysis and finite element modeling of a marimba bar," Acta Acust. united Acust. **85**(2), 258–266 (1999).
[22]W. S. Slaughter, *The Linearized Theory of Elasticity* (Springer Science + Business Media, LLC, New York, 2001).
[23]Mathworks, "Optimization toolbox," https://www.mathworks.com/help/optim/ug/fmincon.html#d123e86976 (Last viewed September 28, 2021).
[24]J. Nocedal and S. Wright, *Numerical Optimization* (Springer, New York, 1999).
[25]H. Gutmann, "A radial basis function method for global optimization," J. Global Optim. **19**, 201–227 (2001).
[26]R. Regis and C. Shoemaker, "A stochastic radial basis function method for the global optimization of expensive functions," INFORMS J. Comput. **19**(4), 497–509 (2007).
[27]Y. Wang and C. Shoemaker, "A general stochastic algorithm framework for minimizing expensive black box objective functions based on surrogate models and sensitivity analysis," arXiv:1410.6271 (2015).
[28]M. Powell, "The theory of radial basis function approximation in 1990," Adv. Numer. Anal. **2**, 105–210 (1992).
[29]MathWorks, "Global optimization toolbox," https://www.mathworks.com/help/gads/surrogateopt.html (Last viewed September 28, 2021).
[30]C. S. Krishnamoorthy, *Finite Element Analysis: Theory and Programming* (Tata McGraw-Hill Eduacation, New Delhi, India, 1994).
[31]J. Juang and R. Pappa, "An eigensystem realization algorithm for modal parameter identification and model reduction," J. Guid. Control Dyn. **8**(5), 620–627 (1985).
[32]R. Pappa and K. Elliot, "Consistent mode indicator for the eigensystem realization algorithm," J. Guid. Control Dyn. **16**(5), 852–858 (1993).
[33]G. Leon and H.-L. Chen, "Direct determination of dynamics elastic modulus and Poisson's ratio of Timoshenko rods," Vibration **2**, 157–173 (2019).
[34]R. Pintelon, P. Guillaume, K. Belder, and Y. Rolain, "Measurement of Young's modulus via modal analysis experiments: A system identification approach," IFAC Proc. Vol. **36**(16), 375–380 (2003).
[35]K.-S. Kim, H.-S. Chang, and N. Akhter, "Determination of Poisson's ratio of a beam by time-average ESPI and Euler–Bernoulli equation," Int. J. Precis. Eng. Manuf. **11**(6), 979–982 (2010).
[36]J. Ferguson, H. Lopez, K. Cho, and C. Kim, "Temperature effects on the tensile properties of precipitation-hardened Al-Mg-Cu-Si alloys," Metals **6**(3), 43 (2016).
[37]A. Villuendas, J. Jorba, and A. Roca, "The role of precipitates in the behavior of Young's modulus in aluminum alloys," Metallurg. Mater. Trans. A **45**, 3857–3865 (2014).
[38]Y. Zhang, Y. Yi, S. Huang, and H. He, "Influence of temperature dependent properties of aluminum alloy on evolution of plastic strain and residual stress during quenching process," Metals **7**(6), 228 (2017).
[39]I. Stanimirovic, "Temperature in machining of aluminium alloys," in *Temperature Sensing* (IntechOpen, London, UK, 2018), pp. 71–89.
[40]P. Larrouy-Maestri, " 'I know it when I hear it': On the listeners' perception of mistuning," Music Sci. **1**, 1–17 (2018).
[41]B. Moore, *An Introduction to the Psychology of Hearing* (Emerald, Bingley, UK, 2013).
[42]W. Hartmann, S. McAdams, and B. Smith, "Hearing a mistuned harmonic in an otherwise periodic complex tone," J. Acoust. Soc. Am. **88**(4), 1712–1724 (1990).
[43]B. Moore, R. Peters, and B. Glasberg, "Thresholds for the detection of inharmonicity in complex tones," J. Acoust. Soc. Am. **77**(5), 1861–1867 (1985).
[44]N. Maia and J. Silva, *Theoretical and Experimental Modal Analysis*, Research Studies, Baldock, UK, 2003).
[45]A. Nayfeh and F. Pai, *Linear and Nonlinear Structural Mechanics* (Wiley-VCH, Weinheim, Germany, 2004).